Lecture 10

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1 Coprimality and Divisibility

1.1 Definitions and Corollaries

- **Definition**: Natural numbers a and b are called **coprime** (or relatively prime) if $\gcd(a, b) = 1$.
  - **Examples**:
    * $\gcd(7, 17) = 1$ (Coprime)
    * $\gcd(8, 9) = 1$ (Coprime)
- **Corollary**: Let $a, b \in \mathbb{N}$. Any common factor of a and b also divides $\gcd(a, b)$.

1.1.1 Proof

We know that there exist integers $x, y \in \mathbb{Z}$ such that:

$$\gcd(a, b) = ax + by$$

Let d be a common divisor such that $d \mid a$ and $d \mid b$. Since d divides both terms a and b, it must divide any linear combination of them:

$$d \mid (ax + by) \implies d \mid \gcd(a, b)$$

- **Example**: Consider $a = 64, b = 28$.
  - $\gcd(64, 28) = 4$.
  - The common factors of 64 and 28 are $\{1, 2, 4\}$.
  - Note that 1, 2, and 4 all divide the GCD (4).

1.2 Euclid's Lemma

Theorem: Let $a, b, c \in \mathbb{Z}$.

1. If $\gcd(a, c) = 1$ and $a \mid bc$, then $a \mid b$.
2. If p is a prime number and $p \mid ab$, then $p \mid a$ or $p \mid b$.

1.2.1 Proof of Part 1

Since $\gcd(a, c) = 1$, by Bézout's identity, there exist integers x, y such that:

$$1 = ax + cy$$

Multiply the entire equation by b:

$$b = abx + bcy$$

We analyze the divisibility of the Right Hand Side (RHS) by a:

- $a \mid abx$ (obviously, as it has a factor of a).
- $a \mid bcy$ (since we are given that $a \mid bc$).

Since a divides both terms on the RHS, a must divide the LHS. Therefore, $a \mid b$.

1.2.2 Proof of Part 2

Let p be a prime such that $p \mid ab$. If $p \mid a$, we are done. Suppose $p \nmid a$. Since p is prime, the only divisors of p are 1 and p. Because $p \nmid a$, they share no common factors other than 1.

$$\implies \gcd(a, p) = 1$$

By Part 1 of this theorem (letting $c = p$), if $\gcd(a, p) = 1$ and $p \mid ab$, then $p \mid b$. Thus, $p \mid a$ or $p \mid b$.

2 The Fundamental Theorem of Arithmetic

2.1 Statement

1. **Existence**: Any natural number $n \geq 2$ can be represented as a product of prime numbers (prime factorization).

2. **Uniqueness**: This representation is unique, up to the ordering of the factors.

2.2 Proof of Existence

We use **Strong Induction** on n.

- **Base Case**: $n = 2$. 2 is prime, so it is a product of primes (itself).

- **Inductive Step**: Suppose the theorem holds for all integers k such that $2 \leq k \leq n$. We must show it holds for $n + 1$.

  - **Case 1**: $n+1$ is prime. Then $n+1 = (n+1)$ is the factorization. Done.
  - **Case 2**: $n + 1$ is composite. Then there exist integers a, b such that:

    $$n + 1 = a \cdot b \quad \text{with} \quad 2 \leq a, b \leq n$$

    By the inductive hypothesis, both a and b can be written as products of primes.

    $$a = \prod p_i, \quad b = \prod q_j$$

    Therefore, $n + 1 = (\prod p_i)(\prod q_j)$ is a product of primes.

2.3 Proof of Uniqueness

Suppose, for the sake of contradiction, that the factorization is not unique. Let n be an integer with two different prime factorizations:

$$n = p_1 p_2 \dots p_k = q_1 q_2 \dots q_l$$

Assume the primes are ordered such that $p_1 \leq p_2 \leq \dots \leq p_k$ and $q_1 \leq q_2 \leq \dots \leq q_l$.

Cancel all common primes from both sides. We obtain a reduced equation:

$$p_{i_1} \dots p_{i_m} = q_{j_1} \dots q_{j_s}$$

where the set of remaining p's and q's are disjoint (no common primes).

Consider p_{i_1} (the first prime on the LHS).

$$p_{i_1} \mid p_{i_1} \dots p_{i_m} \implies p_{i_1} \mid q_{j_1} \dots q_{j_s}$$

By Euclid's Lemma (extended to multiple factors), if a prime divides a product, it must divide at least one of the factors.

$$\implies p_{i_1} \mid q_{j_r} \quad \text{for some } r$$

Since q_{j_r} is also a prime, the only way p_{i_1} can divide it is if:

$$p_{i_1} = q_{j_r}$$

This contradicts the fact that we assumed we had already canceled all common primes. Therefore, the factorization must be unique.

2.4 Canonical Representation

Based on the Fundamental Theorem, we can write any natural number n uniquely as:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

where $p_1 < p_2 < \dots < p_k$ are primes and $\alpha_i \in \mathbb{N}$.